Topic: CO1 Security Features in Real-World Applications

1.Identify any five real-world applications or online services that involve communication, authentication, or transactions. For each application, investigate and identify: The security feature/service required, such as confidentiality, integrity, authentication, authorization, availability The security mechanism used to provide the identified security feature. The security protocol or technology used to implement the mechanism. Briefly explain how the mechanism/protocol provides the identified security feature. Present your findings in the following format:

Application	Security Feature	Security Mechanism	Protocol / Technology	How it provides security
Gmail	Authentication	Password + MFA	OAuth	Verifies the identity of the user
Online banking	Confidentiality	Encryption	HTTPS/TLS	Protects data transmitted between client and server
Adhaar	Authentication	Biometric verification	OTP + Fingerprints/ Iris scan	Verifies identity through unique physical traits
College Result portal	Integrity	Hashing/ Checksum	HTTPS/TLS	Ensures marks are not tampered with in transit
Netflix	Availability	Load balancing, CDN	Content delivery network	Ensures reliable performance under high traffic

Application	Security Feature	Security Mechanism	Protocol / Technology	How it provides security
Instagram	Authentication	Password + OTP	OAut HTTPS	Verifies the identity before granting the account access
Uber/Ola	Authentication	OTP	SMS OTP/ HTTPS	Verifies the driver and rider identity befor the trip

Gmail:

Gmail checks two things before allowing to login. The password and a code sent to the user's phone. This makes sure it's really the user, not someone else pretending to be the user.

Online Banking:

When we send money or check our balance, our data gets encrypted while it travels to the bank's server. So even if a hacker catches it midway, they cannot read it.

Aadhaar:

Aadhaar checks our fingerprint, eye scan, or a phone OTP. These are things only we have, so nobody can fake being us.

College Result Portal:

The marks get a special digital "stamp" when uploaded. If anyone tries to change even one number, the stamp breaks and the system knows something is wrong.

Netflix:

Netflix uses many servers spread across different places. So, when lots of people watch at once, the load gets shared and nothing crashes or lags.

Instagram:

Same as Gmail, password first, then an OTP. This double-check makes sure only we can open our account.

Uber/Ola:

Before our ride starts, we get an OTP. We give it to the driver to confirm. Where, the driver can confirm this as the right ride and the right person.

2.For any five real-world applications, investigate the security controls used to protect users and data. Classify the controls based on whether they provide Confidentiality, Integrity, or Availability, and determine how these controls help in preventing, detecting, and recovering from security attacks. Mention the mechanisms and protocols involved.

Application	Security control	CIA	Preventing	Decteting	Recovering	Mechanism /Protocol
Online banking	Encryption + OTP	Confidentiality	Stops anyone other than users from reading your data or logging in without OTP	Alerts if login is tried from a new device	Allows to freeze account/reset password if hacked	SSL/TLS, HTTPS, OTP, 2FA
Gmail	Spam/Phishing filters + Digital Signatures	Integrity	Blocks fake/malicious emails before they reach inbox	Flags suspicious sender addresses and links	Allows to report phishing, which improves future filtering	DKIM, SPF, DMARC
Amazon	Firewalls + Load Balancers	Availability	Blocks malicious traffic (like	Amazon constantly monit	Redirects traffic to backup	Firewall, DDoS protection, Load Balancing

Application	Security control	CIA	Preventing	Decteting	Recovering	Mechanism /Protocol
			DDoS attacks) from crashing the site	ors traffic patterns.	servers to keep site running	
Instagram	Two-Factor Authentication + Hashing (passwords)	Confidentiality + Integrity	Even if the password leaks in a data breach, the hacker still cannot login without the 2FA code.	Detects login attempts from unknown locations	Allows to recover account via email/phone verification	2FA, Password Hashing (bcrypt), HTTPS
Alexa	Device Authentication + Firmware Updates	Confidentiality + Availability	Stops unauthorized devices from connecting to the home network /hub	Detects unusual command patterns	Auto-updates firmware to patch vulnerabilities and the device daacan factory reset if compromised	Device Pairing Protocols, OTA Updates, WPA3 Wi-Fi Security

Online Banking:

When we login to our bank's app, two things are working hard to keep us safe. First, encryption (SSL/TLS, HTTPS) scrambles our data so that even if someone intercepts it while it is traveling over the internet, all they see is gibberish. This protects our confidentiality. Second, the OTP (one-time password) sent to our phone means that even if someone steals our password, they still cannot get in without that extra code. If someone does try logging in from a device that we have never used before, the bank notices and sends us an alert. That is how it detects something unusual happening. And if our account really gets compromised, we can freeze it or reset our password right away, which is how we recover and stop further damage.

Gmail:

Why most scam emails do not even make it to our inbox? That is due to the spam and phishing filters working alongside protocols like DKIM, SPF, and DMARC. These check whether an email actually came from who it claims to be from, which protects the integrity of our inbox by blocking fake or tampered messages before we even see them.

When something suspicious slips through, Gmail flags it highlighting weird sender addresses or shady links so you know to be careful. And if you do get fooled or spot something fishy, reporting it as phishing helps Gmail's filters get smarter, so it's better at catching similar scams for everyone in the future.

Amazon:

Suppose millions of people are trying to shop on Amazon during a big sale, keeping the site running smoothly for everyone is all about availability. Firewalls and load balancers work together to block harmful traffic (like DDoS attacks meant to crash the site) while spreading normal traffic evenly across servers so that nothing gets overwhelmed. Amazon also keeps a constant eye on traffic patterns.

Instagram:

Passwords get leaked in data breaches more often than the people realize, so Instagram does not rely on passwords alone. It hashes passwords (using something like bcrypt) so that even if a database is stolen, the actual passwords are not readable. On top of that, Two-Factor Authentication (2FA) means that even with a leaked password, a hacker still needs a code sent to your phone or email to actually get in, stopping account takeovers before they even happen. This protects both confidentiality (our password stays private) and integrity (our account cannot be tampered with by someone pretending to be us). If someone tries logging in from an unfamiliar location, Instagram notices and can prompt extra verification. And if our account is ever compromised, we can recover it by verifying our identity through our registered email or phone number.

Alexa:

Since Alexa is always listening for its wake word, keeping unauthorized devices off our home network really matters. Device pairing protocols and WPA3 Wi-Fi security make sure only trusted devices can connect to our Alexa hub, protecting our confidentiality by keeping strangers out of our smart home setup. Alexa also watches for weird behavior, like the device randomly recording without being triggered, which could signal that it is been hacked. If a vulnerability is discovered, Amazon can push out automatic firmware updates over the air to patch the issue and if the device is ever seriously compromised, doing a factory reset wipes it clean, restoring both its security and its normal availability.